SEC 7 · LEC 3 · RADIOISOTOPE GENERATORS (RTG)

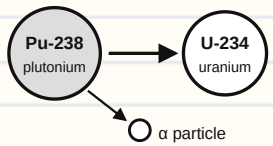
The rock that stays warm

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

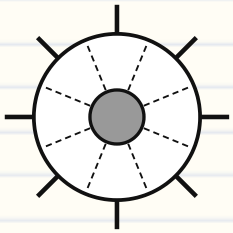
S7 / L3 — RTGs **date · pg __**

1 · A LUMP THAT HEATS ITSELF (decay)

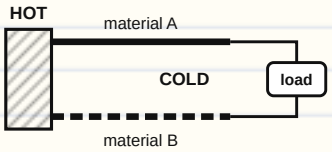


α particle = 2 protons + 2 neutrons
it stops inside the fuel → **HEAT**
1 gram gives 0.57 W of heat
half the atoms gone in 87.7 years
nothing to switch on or off

2 · HEAT → ELECTRICITY (Seebeck effect)



hot fuel in the middle
cold shell with fins



hot end + cold end → a voltage
only about 6% becomes electricity
the rest leaves through the fins
no moving parts

3 · IT FADES SLOWLY

Galileo: 290 W → 250 W in 14 years



4 · WHICH POWER WHERE

SOLAR ARRAY near the Sun, out to Jupiter
BATTERY through the shadow
RTG far from the Sun, or dark for weeks (Pluto, Mars night)

an RTG is not a reactor and not an engine

✏️ **PENCIL KEY** · big circle = atom · small circle = α particle · grey = hot · solid bar = material A · dashed bar = material B · curve = power left

⭐ **3 TAKEAWAYS** 3 lines

1. Plutonium-238 decays by itself and the decay makes heat.
2. Hot inside, cold outside: thermocouples turn about 6% into electricity.
3. Works anywhere, for decades. Used where sunlight is too weak.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What does RTG stand for?	RTG = radioisotope thermal (thermoelectric) generator
What do Voyager 1 and New Horizons have in common?	Voyager 1 + New Horizons: no solar panels → very far from the Sun → RTGs
Where does the heat come from?	heat from the natural decay of plutonium-238 → uranium-234 + an alpha particle (2 protons + 2 neutrons)
What does Pu-238 decay into?	
How does heat become electricity?	Seebeck effect: two different materials, one end hot, one end cold → a voltage
What are the fins for?	hot centre, cold outside · fins keep the outside cold
How long does it last?	power does not depend on the Sun
Why is it controversial?	lasts decades (slide: up to 100 years) · fades as the plutonium decays
Is it a nuclear reactor?	expensive · plutonium is controversial · NASA uses it, ESA rarely
Galileo's RTG: the numbers?	NOT a nuclear reactor: natural decay, no fission chain
? why plutonium-238 and not another	Galileo's: 290 W new → 250 W at end · 55 kg · 7.5 kg Pu-238 · 114 cm long

SEC 7 · LEC 3 · TEACH

Power for where the Sun doesn't reach

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Imagine a lump of metal that is hot to the touch and stays hot for a hundred years with nothing plugged in. Plutonium-238 is that lump. Its atoms are unstable: one by one, each spits out a tiny piece of itself, and every spit is a little kick of heat. A fresh pellet glows red by itself. Nobody starts it and nobody can stop it.

Now the trick for getting electricity. Join two different materials, heat one end and keep the other end cold, and a small voltage appears between the cold ends. So you wrap hundreds of these little pairs around the hot lump and put cooling fins on the outside. That is the whole machine: no moving parts, nothing to wear out. It works in the dark, in a dust storm, at Pluto. The catches: only about 6% of the heat becomes electricity, the plutonium is rare and radioactive so launching it is a political fight, and the power slowly fades as the lump uses itself up. Voyager has been running on three of them since 1977.

🏷 The official names (name first, then what it means)

- **Radioisotope thermoelectric generator, RTG** (the slide's "thermal generator"; nuclear battery): makes electricity from the heat of radioactive decay.
- **Radioisotope** (radioactive isotope): a version of an element whose atoms are unstable. **Radioactive decay**: an unstable nucleus changing by itself and releasing energy.
- **Alpha decay**: decay by throwing out an **alpha particle** (α; a helium nucleus: 2 protons + 2 neutrons).
- **Half-life**: the time for half the atoms to decay. Plutonium-238: 87.7 years.
- **Plutonium-238, Pu-238**: the usual fuel, used as a ceramic, plutonium dioxide. It decays to **uranium-234**.
- **Seebeck effect** (thermoelectric effect): a temperature difference across two joined materials produces a voltage.
- **Thermocouple** (in an RTG, a **unicouple**): one such pair. **Hot junction**, **cold junction**. Material: silicon-germanium.
- **General purpose heat source, GPHS**: the protective brick that holds the fuel pellets.
- **Radiator fins**: carry the waste heat away so the outside stays cold.
- **Thermal power** (heat, watts-thermal) versus **electrical power** (watts-electric). **BOL / EOL**: beginning of life / end of life.
- **Nuclear reactor**: a machine that runs a controlled **fission chain reaction**. An RTG is not one.
- **Radioisotope heater unit, RHU**: a small pellet used only to keep equipment warm.

🛠 How it actually works

Step	The mechanism, with numbers (Galileo's RTG, from the slide)
1 · Decay	A plutonium-238 nucleus (94 protons, 144 neutrons) throws out an alpha particle and becomes uranium-234 (92 and 142). Count them: 94 − 2, 144 − 2.
2 · Heat	The alpha particle is stopped inside the fuel within a hair's width, and its energy turns into heat. One gram gives 0.57 W. Galileo's 7.5 kg: about **4,300 W** of heat. The fuel sits near 1,000 °C.
3 · Seebeck	Heat makes the charge carriers in a material drift from the hot end to the cold end. Two different materials drift by different amounts, so joining them at the hot end leaves a voltage between the cold ends. 572 pairs in series give about 30 V.
4 · Efficiency	Hot side about 1,000 °C, cold side about 300 °C. 290 W of electricity from 4,300 W of heat: **6.7%**. The other 4,000 W leaves through the fins (and helps keep the spacecraft warm).
5 · Fading	Power = starting power × 0.5^(years ÷ 87.7). That is 0.8% a year. Galileo flew 14 years: 0.5^(14 ÷ 87.7) = 0.895, so 290 W → 260 W from decay alone. The thermocouples age too, which gives the slide's **250 W**.
6 · Voyager	470 W at launch in 1977. About 220 W today, as far as I know, so instruments are being switched off one at a time.
7 · Why this isotope	Alpha particles are stopped by a sheet of paper, so almost no shielding is needed. And 88 years is long enough for a mission but short enough to be hot. Strontium-90 (29 years) needs heavy shielding. Americium-241 (432 years) gives a fifth of the heat per gram.
8 · Safety	The fuel is ceramic pellets clad in iridium inside graphite shells built to survive a launch accident and re-entry. Apollo 13's is lying intact on the floor of the Pacific. Plutonium-238 is not bomb material; that is plutonium-239.

SEC 7 · LEC 3 · COMPARE + FACT CHECK

Three ways to keep the lights on

Section 7 in one table, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

📋 The power sources side by side

Source	Energy comes from	Typical power	Lasts	Use it when	The catch
Solar array	sunlight	100s of W to 100,000+ W	15+ years, fading 1–3% a year	anywhere from Mercury to Jupiter	nothing in shadow; huge beyond Mars
Secondary battery	stored from the array	the whole spacecraft, for an hour	thousands of cycles	every eclipse	stores, does not make; heavy
Primary battery	its own chemicals, once	W to kW for hours	one use	pyro, short-lived probes	cannot be recharged
RTG	decay heat of plutonium-238	100–300 W each	decades, fading 0.8% a year	beyond Jupiter; long nights; dust	6% efficient; costly; scarce fuel; politics

RTG users: Voyager 1 and 2, Galileo, Cassini, New Horizons, and the Mars rovers Curiosity and Perseverance (for the dust and the nights, not the distance).

▲ Where the instructor was loose

- **The name.** His slides say radioisotope *thermal* generator. The proper name is radioisotope **thermoelectric** generator.
- **"They can last up to 500 years."** His own slide says 100. The honest answer: the heat halves every 87.7 years and the thermocouples age as well, so useful life is decades. Voyager is at 49 years and running short.
- **"One point fifty metres long."** His slide says 114 cm.
- **"Temperature is ejected from the nucleus."** A particle is ejected. It is stopped inside the fuel and its energy of motion becomes heat.
- **"Really only applicable for space."** The Soviet Union powered hundreds of remote lighthouses with strontium-90 units, and early heart pacemakers ran on plutonium-238.
- **"ESA is developing strontium."** As far as I know, the European programme is built on americium-241. Strontium-90 was the Soviet ground choice.
- **Only for "very far from the Sun".** Also for long darkness and dust: the Mars rovers, and the Apollo instruments that had to survive two-week lunar nights.
- **"New Horizons took a few poor photos of Pluto."** Probably the captions. It returned sharp close-ups of Pluto in 2015.
- **Not said: the efficiency (about 6%)**, and that an RTG cannot be turned off. It is hot on the launch pad and has to be cooled there.
- **He was right** that it is not a reactor: no chain reaction, no fission, nothing to control. And this is the "RTG is not an engine" point from Section 4.

🔎 Caption errors

Captions said	Instructor meant
"radio isotope with more generators"	radioisotope **thermal** generators
"apart from the intent"	apart from the **antenna**
"far apart from soccer"	far from the **Sun**
"the five to one"	**Voyager 1**
"plutonium to 3:8"	plutonium-**238**
"the case time to uranium… not a particle"	**decays into** uranium… **an alpha** particle
"to newsrooms and to printers"	**two neutrons and two protons**
"X thermic"	**exothermic** (gives off heat)
"Seabeck", "seabag", "C-PAC effect"	**Seebeck** effect
"occult material", "Hunter called Hellmann's"	**a cold** material, **hot and cold elements**
"a strong team"	**strontium**
"loan to date on RTG needs"	**long debate** on RTGs
"regarding Cephalopoda"	regarding **RTGs**

🗣 Say it out loud

Friend: "How does Voyager still have power after fifty years with no solar panels?"

You: It carries lumps of plutonium-238, which is naturally hot because its atoms keep breaking down. Around the hot lump are hundreds of little pairs of two materials: hot on one end, cold on the other, and that difference makes a voltage. No moving parts, works in total darkness. It is not a reactor, just natural decay. It only turns about 6% of the heat into electricity and it loses a bit under 1% a year, which is why Voyager is now switching instruments off.

SEC 7 · LEC 3 · VOCABULARY

Every word used, in plain English

32 entries. "Repeat" = you have met it before.

Radioisotope thermoelectric generator (RTG) — a device that turns the heat of radioactive decay into electricity. *A rock that stays warm, wrapped in heat-to-electricity strips.*

Radioisotope — a form of an element whose atoms are unstable and break down by themselves.

Isotope — one version of an element; versions differ in their number of neutrons.

Radioactive decay / natural decay — an unstable atom changing into another by itself, releasing energy.

Spontaneous — happening by itself, with no trigger.

Nucleus — the tiny heavy centre of an atom, made of protons and neutrons.

Proton / neutron — the positive / neutral particles in the nucleus.

Plutonium-238 (Pu-238) — the radioisotope used as RTG fuel. Half-life 87.7 years.

Uranium-234 (U-234) — what plutonium-238 turns into.

Alpha particle (α) — two protons and two neutrons thrown out together: a helium nucleus. *Stopped by a sheet of paper.*

Half-life — the time it takes for half the atoms to decay.

Exothermic — giving off heat.

Seebeck effect (thermoelectric effect) — a voltage produced by a temperature difference across two joined materials.

Thermocouple / junction — a pair of different materials joined together / the place they join.

Hot / cold side — the end against the fuel / the end against the finned shell.

Fins (radiator fins) — thin plates that throw away waste heat so the outside stays cold. *Repeat idea, Section 2.*

Voltage — electrical push. *Repeat.*

Watt (W) — unit of power. *Repeat.*

BOL / EOL — beginning of life / end of life: power when new and when the mission ends.

Interplanetary / interstellar — between planets / between stars. *Repeat.*

Controversial — argued over. *Repeat.*

Strontium-90 / americium-241 — other radioisotopes that can fuel such generators.

Nuclear reactor — a machine that splits atoms in a controlled chain reaction. *Repeat, Section 4.*

Fission / fusion — splitting heavy atoms / joining light ones. An RTG does neither.

Chain reaction — each split atom triggering more splits.

Shielding — material placed to stop radiation.

Heat source (GPHS) — the protective brick holding the plutonium pellets.

Multi-foil insulation — layers of thin foil that keep the heat going through the thermocouples. *Like MLI, Section 2.*

Voyager 1 — launched 1977; now beyond the Sun's bubble, in interstellar space. *Repeat, Who's Who.*

New Horizons — the Pluto probe; one RTG. *Repeat, Who's Who.*

Galileo — NASA's Jupiter orbiter of the 1990s; two RTGs. *Repeat, Who's Who.*

NASA / ESA — the US and European space agencies. *Repeat.*